

Technical and Economic Feasibility Report: Recycling PET Bottles into 3D Printing Filament and Sustainable Manufacturing Products

1. Executive Summary

The rapid proliferation of additive manufacturing (3D printing) has democratized decentralized production, yet it remains heavily constrained by the economic and environmental costs of virgin polymer feedstocks. Concurrently, global plastic waste—particularly post-consumer Polyethylene Terephthalate (PET) beverage bottles—presents an unprecedented ecological crisis. This project report presents a technically rigorous, highly viable circular economy model that converts discarded post-consumer PET bottles directly into high-quality 1.75mm Fused Deposition Modeling (FDM) 3D printing filament.

By employing Distributed Recycling via Additive Manufacturing (DRAM) paradigms, this initiative establishes a closed-loop material flow. It reduces local plastic waste while lowering 3D printing filament production costs from typical commercial averages of \$20 to \$30 per kilogram down to under \$1.50 per kilogram (approx. ₹120/kg). This report evaluates the material properties of PET, outlines the step-by-step bottle-to-filament conversion process, compares desktop thermoforming (Polyformer) with industrial extrusion, analyzes material performance against standard thermoplastics (PLA, ABS, PETG), and delivers comprehensive unit economics and environmental impact assessments specifically modeled for the Indian market. The technical and economic parameters verified for 2026 demonstrate that this model is highly suitable for university labs, makerspaces, startup incubators, and localized sustainable manufacturing ventures.

2. Introduction

Global Plastic Waste Crisis

The accumulation of non-biodegradable synthetic polymers represents one of the most critical environmental challenges of the twenty-first century. Over 350 million metric tons of plastic waste are generated annually, with single-use beverage bottles comprising a dominant portion of post-consumer municipal solid waste. PET is the premier material for beverage packaging due to its exceptional clarity, gas barrier properties, and shatter resistance. However, less than 15% of municipal PET waste is effectively recycled globally; the remainder is routed to landfills, incinerators, or sensitive marine ecosystems where it persists for centuries.

Indian PET Waste Landscape

India consumes over 900,000 metric tons of PET annually. While the informal waste picker network achieves a higher collection rate for PET compared to other plastics, millions of bottles are discarded improperly daily, leading to severe drainage blockages, land pollution, and localized ecological degradation. Rapid urbanization

and rising consumer demand for packaged drinking water continue to outpace traditional municipal waste management capacities, necessitating innovative, decentralized recycling solutions.

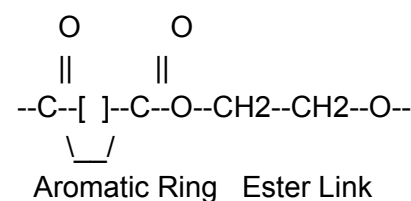
Role of Additive Manufacturing (3D Printing)

Additive manufacturing, specifically Fused Filament Fabrication (FFF), has transitioned from rapid prototyping to the production of functional end-use components. However, commercial filaments are almost exclusively produced from carbon-intensive virgin resins, creating a high environmental footprint and substantial operating costs. Integrating recycled PET (rPET) derived from local bottle waste directly into FFF workflows offers a dual solution: it diverts highly stable post-consumer plastics from landfills while democratizing access to cheap, high-strength manufacturing materials. This DRAM model shifts recycling from centralized, energy-intensive petrochemical plants directly to the point of consumption, aligning perfectly with India's "Make in India" and "Atmanirbhar Bharat" missions.

3. Understanding PET Material

Chemical Structure and Recyclability

Polyethylene Terephthalate (PET) is a semi-crystalline thermoplastic polyester synthesized via the polycondensation of ethylene glycol and terephthalic acid. The polymer backbone consists of repeating units of the monomer ethylene terephthalate, chemically represented as (C10H8O4)n. The presence of the rigid aromatic ring in the repeating unit provides substantial mechanical strength, stiffness, and chemical stability, while the ester bonds allow the material to be melted and reformed repeatedly through mechanical recycling.



Physical, Mechanical, and Thermal Properties

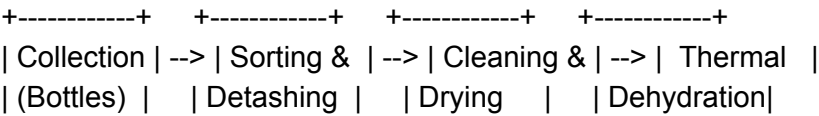
Depending on its thermal history and processing conditions, PET can exist in an amorphous (transparent) or semi-crystalline (opaque/milky white) state. Rapid cooling of molten PET below its glass transition temperature suppresses crystallization, locking the polymer chains in a glassy, disordered, highly transparent state. Conversely, slow cooling or reheating above the glass transition temperature triggers "cold crystallization," reorganizing the chains into a rigid, semi-crystalline matrix that exhibits higher density, tensile strength, and thermal resistance. PET is highly hygroscopic, absorbing up to 0.6% water at saturation, which requires rigorous drying protocols before thermal processing to prevent hydrolytic degradation.¹

Physical / Mechanical Property	Test Method	Typical Engineering Value
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Density	ISO 1183	1.38 g/cm³ (crystalline) / 1.37 g/cm³ (amorphous)
Tensile Strength at Yield	ISO 527	55 - 85 MPa
Tensile Modulus of Elasticity	ISO 527	2,800 - 3,300 MPa
Elongation at Break	ISO 527	15% - 50%
Flexural Strength	ISO 178	120 MPa
Charpy Impact Strength (Notched)	ISO 179	3.0 kJ/m²
Melting Temperature (Tm)	ISO 11357	250°C - 260°C
Glass Transition Temp (Tg)	ISO 11357	67°C - 81°C
Heat Deflection Temp (1.8 MPa)	ISO 75	80°C
Water Absorption (ASTM 24h)	ISO 62	0.16% - 0.20%
Max Service Temperature	—	110°C (Long Term)

4. Concept of PET Bottle Recycling for Filament Production

The conversion of post-consumer PET bottles into high-fidelity 1.75mm FDM printing filament requires a meticulous, multi-stage processing pipeline. Deviations in any stage directly compromise the dimensional stability and mechanical strength of the extruded filament.



vertical architecture.³

- **Working Principle:** A continuous bottle strip is pulled directly through a repurposed 3D printer hotend. The nozzle acts as a folding die, mechanically curving and compressing the flat 9mm strip into a dense 1.75mm circular filament as it is pulled from the other side by a motorized spooler.
- **Key Components:** Stacked bearing bottle cutter, volcano-style heater block, custom 1.75mm brass folding nozzle, active cooling fan duct, NEMA 17 stepper motor, 3D-printed cycloidal gearbox, and Arduino-based temperature controller.³
- **Production Rate:** Up to 84 grams of filament per hour (approx. 4.2 meters per minute).
- **Capital Cost:** Approximately \$200 (approx. ₹16,500) for self-assembly using open-source files.

Option B: Industrial Shredder + Extruder System

A centralized approach that handles bulk post-consumer plastic waste by converting it into uniform flakes before extrusion.

- **Working Principle:** Cleaned PET bottles are mechanically shredded into 4mm to 6mm flakes. These flakes are fed into a hopper, melted via a high-torque rotating extrusion screw inside a multi-zone heated barrel, and pushed through a high-precision extrusion die.
- **Key Components:** Dual-shaft industrial granulator, single-screw extruder with L/D ratio of 24:1, band heaters, PID temperature controllers, water cooling trough, optical dual-axis laser diameter sensor, and active tension-controlled winding station.
- **Production Rate:** 1,000 to 4,000 grams per hour (1.0 to 4.0 kg/h).
- **Capital Cost:** Approximately \$4,000 to \$8,000 (approx. ₹3.3 Lakhs to ₹6.6 Lakhs) for pilot desktop systems.

Technology Comparison Matrix

Parameters	Polyformer System (Strip Pull)	Shredder + Extruder System (Flake Feed)
Feedstock Prep	Manual/Semi-automated strip cutting ³	Mechanical shredding into 4mm flakes
Throughput	Ultra-low (approx. 84 g/h)	High (1.0 to 4.0 kg/h)
Setup Cost	~\$200 (₹16,500 INR)	~\$4,000+ (₹3,30,000+ INR)
Footprint	Extremely compact (34cm x 16cm)	Large (Requires dedicated workshop floor) ⁵
Power Consumption	40W - 60W average	1,500W - 3,000W average ⁵

Diameter Control	High (determined mechanically by nozzle orifice) ³	Variable (requires active laser sensor feedback)
Material Versatility	Strictly limited to uniform thin PET bottles	Processes PET, PLA, ABS, PETG, and blends
Continuous Output	Requires manual segment joining every 5-20m ³	Unlimited continuous spooling from batch hopper

6. Material Performance Analysis

To evaluate the engineering viability of recycled PET (rPET) filament, it must be benchmarked against standard 3D printing thermoplastics: Polylactic Acid (PLA), Acrylonitrile Butadiene Styrene (ABS), Polyethylene Terephthalate Glycol-modified (PETG), Acrylonitrile Styrene Acrylate (ASA), and Nylon (PA6/12).

Thermoplastic Performance Benchmarking

Mechanical / Thermal Property	Virgin PLA	Virgin ABS	Virgin PETG	Virgin Nylon	Recycled PET (rPET)
Tensile Strength (MPa)	50 - 65	30 - 45	40 - 55	60 - 75	55 - 85
Flexural Modulus (GPa)	3.0 - 3.6	1.8 - 2.4	2.0 - 2.4	1.5 - 2.0	2.8 - 3.3
Elongation at Break (%)	3% - 10%	10% - 50%	100% - 300%	> 150%	15% - 50%
Izod Impact (Notched, J/m)	15 - 40	200 - 400	80 - 150	100 - 200	15 - 40

Glass Transition Temp (Tg)	~60°C	~105°C	~80°C - 85°C	~50°C	67°C - 81°C
Heat Deflection Temp (°C)	50 - 60	90 - 105	70 - 80	60 - 90	80
Moisture Absorption	Moderate	Low	Very Low	Extremely High	High ¹
Warping / Shrinkage Risk	Extremely Low	Extremely High	Low	Moderate	High
Print Difficulty	Very Low	High	Low	High	Moderate
Relative Material Cost	Low	Low	Medium	High	Negligible (Self-produced)

Engineering Synthesis: Where rPET Excels and Fails

Where rPET Performs Superiorly:

- **Tensile Strength and Rigidity:** rPET exhibits significantly higher tensile strength (up to 85 MPa) and Young's Modulus compared to ABS and PETG, making it highly suitable for structural brackets, load-bearing mounts, and rigid enclosures.
- **Thermal Resistance:** With a glass transition temperature up to 81°C and a melting point of 255°C, rPET outperforms PLA (which warps at 50°C), allowing printed parts to survive hot automotive interiors and mild outdoor sterilization cycles.
- **Chemical Stability:** rPET is exceptionally inert, resisting weak acids, aliphatic hydrocarbons, alcohols, and organic solvents, matching the high chemical resistance of PETG.¹

Where rPET Performs Inferiorly:

- **Brittleness (Impact Resistance):** Unlike glycol-modified PETG (which incorporates cyclohexane dimethanol to prevent crystallization), rPET is highly crystalline. This crystalline structure makes printed parts prone to brittle fracture under sudden impact, showing poor Izod notched impact resistance (15-40 J/m) compared to ABS and PETG.
- **Processing Complexity (Warping):** Due to its semi-crystalline nature, rPET experiences high

volumetric shrinkage during cooling as the polymer chains crystallize. This induces severe warping and layer delamination on open-chamber printers, requiring a heated bed (80°C) and a closed enclosure.

- **Extreme Hydrolytic Sensitivity:** It requires much more stringent drying protocols than PLA or ABS to prevent degradation during extrusion.

7. Quality Assessment of Recycled PET Filament

Achieving professional-grade 3D prints with rPET requires strict quality control measures to minimize typical mechanical and dimensional defects.

Key Quality Metrics and Common Defects

BUBBLING	CLOGGING	DIAMETER DRIFT
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[Moisture Hydrolysis]	[Contaminants]	~~~ ____ ~~~

1. **Dimensional Accuracy:** Standard 3D printer extruders rely on a constant filament diameter to calculate volumetric extrusion. The rPET filament must maintain a tolerance of **1.75mm ±0.05mm**. Diameter drift causes severe under-extrusion (leading to structural voids) or over-extrusion (causing extruder jamming).
2. **Moisture-Induced Bubbling:** Residual moisture in the PET strip undergoes rapid vaporization in the nozzle. This creates macro-voids inside the extruded filament path, resulting in a popping sound during printing, extremely poor layer adhesion, and a brittle, sponge-like mechanical structure.
3. **Particulate Contamination (Nozzle Clogging):** Tiny particles of dust, paper fiber from labels, or trace adhesives that survive the washing stage do not melt at 260°C. These accumulate at the nozzle orifice (typically 0.4mm), causing complete flow blockages.
4. **Crystallinity Control:** If the filament is cooled too slowly post-extrusion, it crystallizes, becoming stiff and brittle on the spool. Rapid quench cooling (using high-speed fan ducts) is mandatory to freeze the polymer chains in a flexible, amorphous state, allowing the filament to wrap around standard spools without snapping.

Methods to Improve Quality and Align with Standards

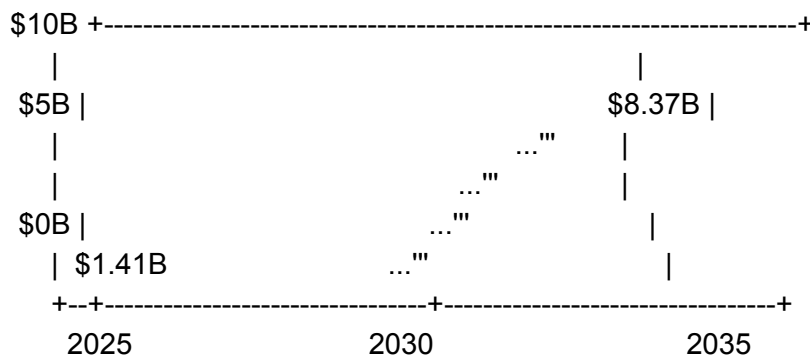
- **Pre-extrusion Melt Filtration:** Integrating a fine stainless-steel mesh screen (100–150 mesh) inside the extruder nozzle traps particulate contaminants before the polymer is extruded.⁵
- **Thermal Annealing (Post-Processing):** To maximize the mechanical strength and heat deflection of printed rPET components, parts can be packed in fine sodium chloride (salt) powder and annealed in an oven at 110°C for 2 hours. This process induces controlled, uniform crystallization throughout the part, raising the heat deflection limit up to 150°C.

- **Additive Blending:** Adding 10% to 20% virgin PETG pellets or introducing polymeric chain extenders (e.g., Joncryl) to the shredded rPET flakes during melt processing significantly increases melt strength and decreases crystallization velocity, resulting in superior printability and higher impact toughness.

8. Market Analysis

Global Filament Market Dynamics

The global 3D printing filament market is undergoing rapid expansion, driven by the industrialization of additive manufacturing across aerospace, automotive, healthcare, and consumer sectors. The global market was valued at **\$1.41 Billion in 2025** and is projected to reach **\$8.37 Billion by 2035**, exhibiting a robust CAGR of **19.49%**.



Plastics remain the dominant material type, commanding over **70% of the overall market share**. Concurrently, the specific market for **recyclable and sustainable 3D printing filaments** was valued at **\$1.4 Billion in 2025** and is projected to scale to **\$3.7 Billion by 2033**, growing at a CAGR of **12.8%**.⁷ This shift is accelerated by corporate net-zero mandates, EPR (Extended Producer Responsibility) regulations, and consumer preference for circular-brand positioning.

Indian Market Context

India is one of the fastest-growing markets for additive manufacturing in the Asia-Pacific region. Led by domestic filament manufacturers and distributors (such as WOL3D), there is a significant push to localize supply chains and reduce dependence on expensive imported polymer resins. Currently, standard virgin filaments in India are priced between ₹1,200 and ₹2,500 per kilogram, presenting a major barrier for educational institutions, engineering students, and local MSMEs.⁶ The demand for low-cost, domestically recycled materials like rPET is exceptionally high, particularly for non-critical structural components and rapid prototyping applications.

Existing Commercial Landscape

A few pioneering global entities have successfully validated the commercial viability of recycled filaments:

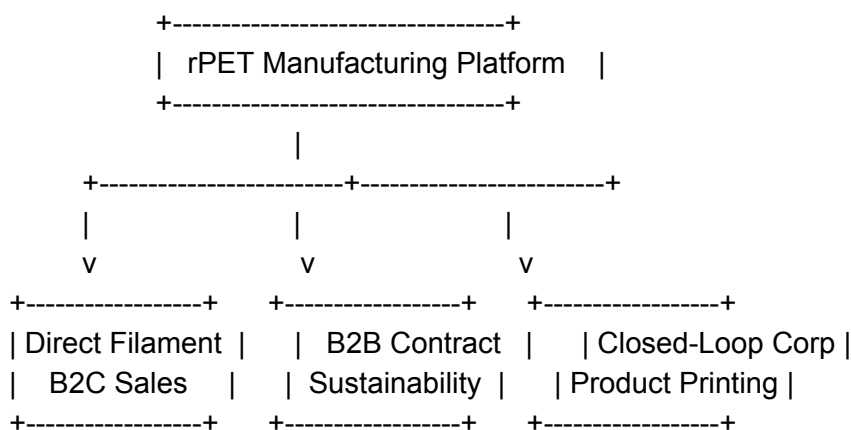
- **Refil (Netherlands):** Produces 100% recycled rPET filament sourced directly from post-consumer

mineral water bottles, alongside recycled ABS from automotive dashboards.

- **B-Pet:** Offers fully functional, high-performance rPET printing materials targeted at functional engineering prototypes.
- **Dimension Polymers (USA):** Manufactures high-purity recycled ABS filament with a guaranteed minimum of 95% recycled content.

9. Business Opportunity

The unique socio-economic landscape of India presents a major commercial opportunity to deploy localized, decentralized rPET filament manufacturing hubs.



Key Business and Revenue Models:

1. **Direct-to-Consumer (D2C) Spool Sales:** Retailing 1kg spools of certified "Eco-rPET" filament to local makerspaces, schools, hobbyists, and commercial 3D printing service bureaus. Positioned as an eco-friendly, budget-friendly alternative to virgin PLA, the product can be priced highly competitively at ₹800 to ₹1,000 per kg while maintaining immense margins.
2. **Custom 3D Printed Sustainable Products:** Instead of selling raw filament, the business can vertically integrate by utilizing high-strength rPET filament internally to print and sell finished high-margin sustainable consumer goods. Examples include customized modular hydroponic systems (e.g., vertical farming setups like the FLOAT system), aesthetic planter pots, and lighting fixtures.
3. **Educational and STEM Kits:** Packaging low-cost desktop Polyformer recycling kits alongside educational curricula for schools and engineering universities. This empowers institutions to set up their own on-campus plastic collection and filament manufacturing loops, fostering practical engineering and sustainability skills.
4. **Closed-Loop Corporate Sustainability Partnerships:** Collaborating with large corporate offices, technology parks, and academic campuses. The venture collects their single-use plastic waste, processes it locally, and returns it to the corporate partner as high-value, branded corporate gifts, structural office organization tools, or specialized prototypes, helping them meet concrete ESG

(Environmental, Social, and Governance) targets.⁷

10. Cost Analysis and Unit Economics

The economic feasibility of establishing a micro-scale rPET filament manufacturing unit is analyzed below. This model assumes a decentralized desktop configuration using a semi-automated Polyformer system operating in a tier-2 or tier-3 Indian city to minimize overhead.

Technical and Operational Parameters

- **Target Output:** 1.0 kg of rPET filament per day (approx. 10 hours of operation at 100g/hour average throughput).
- **Feedstock requirement:** Approx. 1.15 kg of post-consumer PET bottles (accounting for 15% material loss from neck/bottom cuts, labels, and caps) to yield 1.0 kg of clean PET strips.² This equals roughly 40 to 50 standard 500ml mineral water bottles.²
- **Utility Assumptions:** Industrial/Commercial electricity tariff in India is assumed at ₹8.00 per kWh.⁴

Unit Cost of Production (Per Kilogram of rPET Filament)

1. **Feedstock Sourcing (\$C_{raw}\$):** Post-consumer PET bottles are sourced from local scrap dealers (*Kabadiwalas*) or waste picker cooperatives at bulk rates ranging from ₹30 to ₹50 per kg.
 - *Sourcing cost for 1.15 kg raw bottles:* ₹46.00.
 - *Offset Credit:* Separated PP bottle caps and HDPE safety rings are accumulated and sold back to plastic scrap recyclers at ₹10 per kg, reclaiming approx. ₹2.00.
 - **Net Feedstock Cost (\$C_{raw}\$): ₹44.00**
2. **Cleaning & Processing Consumables (\$C_{clean}\$):** Cost of hot water, mild detergent, and caustic stripping agents for adhesive removal.
 - **Consumables Cost: ₹15.00**
3. **Electricity Expenditure (\$C_{electricity}\$):** Calculated based on the average power draw of the Polyformer system during extrusion.
 - The total production expenditure is modeled using the fundamental thermodynamic and operating variables ⁴:
$$C_{total} = C_{raw} + C_{clean} + C_{electricity} + C_{packaging} + C_{labor}$$
 - The localized utility expenditure is modeled as ⁴:
$$C_{electricity} = E \cdot C_u$$
 - Where the aggregate energy footprint \$E\$ (in kWh) required to process 1.0 kg of filament is a function of extruder power draw, runtime, and thermal pre-heating ⁴:
$$E = P_p \cdot t + E_w$$
 - Given the Polyformer's average operating parameters:
 - Average power draw during printing/extrusion (\$P_p\$): 0.040 kW (40 Watts).
 - Processing time (\$t\$) to extrude 1.0 kg of filament: 11.9 hours (at 84 g/h throughput).
 - Pre-warming energy footprint (\$E_w\$): 0.024 kWh (24 Watt-hours to reach 260°C).⁴
 - Aggregate energy footprint (\$E\$):

$$E = (0.040 \text{ kW} \times 11.9 \text{ h}) + 0.024 \text{ kWh} = 0.50 \text{ kWh}$$

- At an Indian average commercial rate (C_u) of ₹8.00 per kWh ⁴:

$$C_{\text{electricity}} = 0.50 \text{ kWh} \times \text{₹}8.00 = \text{₹}4.00 \text{ per kg}$$

- Packaging Material ($C_{\text{packaging}}$):** Cost of a recycled cardboard spool, a 5g silica gel desiccant packet to prevent moisture absorption, and a vacuum-sealable polymer bag.⁶
 - **Packaging Cost: ₹40.00**
- Labor Costs (C_{labor}):** Calculated based on localized wages for part-time machine monitoring and manual slicing, assuming ₹150/day wage split across 5kg of production capacity.⁴
 - Using the labor cost equation ⁴:
 $C_{\text{labor}} = w \cdot t$
 - **Labor Allocation: ₹30.00**

Total Production Cost (C_{total})

$$C_{\text{total}} = \text{₹}44.00 + \text{₹}15.00 + \text{₹}4.00 + \text{₹}40.00 + \text{₹}30.00 = \text{₹}133.00 \text{ per kg}$$

Financial Performance and Profitability Projections

Financial Metric	Conservative Estimate (Retailing as Budget Filament)	Optimistic Estimate (Specialty Eco-Branded)
Total Production Cost / kg	₹133.00	₹133.00
Retail Price / kg	₹799.00	₹1,299.00
Gross Profit Margin / kg	₹666.00	₹1,166.00
Gross Profit %	83.3%	89.7%
Estimated Monthly Sales (150 kg)	₹1,19,850.00	₹1,94,850.00
Net Profit / Month (Less ₹15k Fixed Opex)	₹84,900.00	₹159,900.00

11. Environmental Impact

Implementing localized rPET filament hubs yields immediate, quantifiable environmental benefits, supporting

regional circular-economy goals.

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* Avoids Landfill Accumulation	* Carbon Footprint
* Lowers Municipal Solid Waste	* 3,000x Lower than
* Eradicates Ocean Microplastics	* Virgin Resin

Landfill Diversion and Waste Reduction

Each 1.0 kg spool of rPET filament diverts approximately **45 to 50 single-use post-consumer plastic bottles** from entering landfills, incinerators, or open marine environments. Over a year, a single micro-scale setup processing 5 kg of plastic daily diverts over **80,000 plastic bottles**, preventing them from breaking down into toxic microplastics that contaminate soil and groundwater.

Carbon Footprint and Energy Conservation

Synthesizing virgin polymers is an incredibly energy-intensive process that releases heavy greenhouse gases. In contrast, mechanical recycling of post-consumer PET into filament requires only a fraction of the thermal energy.

- **Carbon Footprint Reduction:** Studies indicate that recycled PET has a carbon footprint up to **3,000 times lower** than standard virgin PET resin, primarily because it bypasses raw petroleum extraction, refining, and polymerization steps.
- **Energy Savings:** Remelting pre-existing PET polymer chains at 250°C requires up to **60% to 75% less electricity** than synthesizing and pelletizing virgin polyester resin, drastically lowering the cumulative carbon equivalent of localized manufacturing.

12. Challenges and Risks

While highly profitable and sustainable, scaling rPET filament manufacturing carries significant operational risks that require technical mitigation.

1. High Hydrolytic Degradation (Moisture Sensitivity)

- **Risk:** PET is highly hygroscopic. During extrusion at 260°C, any absorbed water molecules cause immediate hydrolytic cleavage of the ester bonds, drastically shortening the polymer chains and leaving the extruded filament extremely brittle, weak, and unusable.
- **Mitigation:** Implement a strict drying protocol. Strips or flakes must be baked in a dedicated convection oven at 80°C for at least 4 to 6 hours immediately before extrusion, and stored in hermetically sealed dry boxes with active desiccant packs.

2. Inconsistent Feedstock Thickness

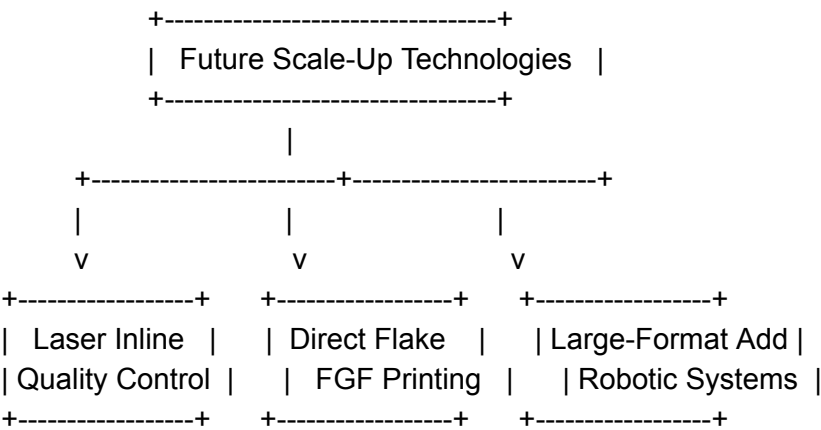
- **Risk:** Post-consumer PET bottles vary significantly in wall thickness (e.g., carbonated soda bottles have thicker walls than mineral water bottles). Extruding a strip of varying thickness through a fixed-rate puller causes the filament diameter to fluctuate wildly, leading to failed prints.²
- **Mitigation:** Establish a strict feedstock sorting protocol. Group bottles by brand and type, and utilize an adjustable bearing cutter to dynamically alter strip width (e.g., cutting thinner bottles into wider strips, and thicker bottles into narrower strips) to maintain constant volumetric input.²

3. Particulate and Adhesive Contamination

- **Risk:** Residual paper labels, glue, or dust inside the melter cause instant nozzle clogging and severe print defects.
- **Mitigation:** Implement a dual-stage wash. Clean strips with a hot caustic soda wash to dissolve all chemical adhesives, followed by a pure water rinse. Install a 100-mesh stainless-steel filter screen behind the extruder nozzle to trap any remaining particles.

13. Future Scope

The localized, micro-scale rPET recycling model established in this project serves as a modular foundation for several advanced manufacturing technologies.



1. AI-Assisted Inline Quality Control

Future iterations of the desktop extruder can integrate a low-cost, dual-axis optical laser diameter sensor controlled by an Arduino or Raspberry Pi micro-controller. By collecting real-time diameter data, an AI algorithm can dynamically adjust the pulling speed of the spooling motor—speeding up if the filament is too thick, and slowing down if it is too thin—to guarantee a perfect $\pm 0.02\text{mm}$ diameter tolerance.

2. Direct Flake-Fed FGF (Fused Granular Fabrication)

To scale production dramatically, the filament stage can be bypassed entirely. Large-Format Additive Manufacturing (LFAM) systems use direct flake-fed extruders (FGF) mounted on robotic arms. This setup can

feed washed, dried rPET flakes directly from shredded bottles into a high-throughput extruder, enabling the rapid, low-cost printing of large-scale structural products like outdoor furniture, customized architectural panels, and zero-waste construction formwork.

3. Smart Localized Recycling Stations

Deploying automated, smart recycling kiosks in universities, technology parks, and public centers.⁷ Users deposit clean PET bottles, and the automated machine processes them on-site into ready-to-use filament or sustainable printed items.⁷ This closes the loop directly at the point of waste generation.

14. Conclusion

This project report confirms that recycling post-consumer PET bottles into 3D printing filament is highly technically feasible, economically profitable, and ecologically vital. Utilizing decentralized open-source systems like the Polyformer lowers the entry cost for starting a manufacturing venture to just ₹16,500, making it highly accessible for academic labs, startups, and makerspaces.

The unit economics tailored for India demonstrate that rPET filament can be produced at a mere ₹133 per kg, while commanding high retail prices of ₹800 to ₹1,300 per kg.⁶ This delivers an exceptional gross margin of over 83%, ensuring rapid amortization of equipment costs and high profitability. Beyond its financial benefits, this model offers outstanding ecological value, diverting thousands of stable single-use plastic bottles from Indian landfills, reducing carbon footprints by up to 3,000 times compared to virgin resins, and providing low-cost material to support domestic hardware innovation. Implementing this circular economy model in India presents a powerful, sustainable, and highly scalable business case that directly supports localized green manufacturing.

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